Pascal's triangle

1 The 4th row of Pascal's triangle consists of the numbers $1, 4, 6, 4, 1$.

a Write down the numbers in the 5th row of Pascal's triangle.

b Find the sum of the entries in the 5th row of Pascal's triangle.

c Rewrite each element of the 5th row of Pascal's triangle as an ${}^{n}C_{r}$ coefficient.

2 Complete the following entries in a particular row of Pascal's triangle:

$$1, 8, \square, 56, 70, \square, 28, \square, 1$$

3 Which row of Pascal's triangle would give us the coefficients of terms in the expansion of $(x + y)^7$?

4 State the number of terms in the following expansions:

a $(4x + y)^6$ **b** $(m + y)^8$

5 Consider the expansion of $(y + k)^4$.

a How many terms will there be in the expansion?

b Write the coefficient of the 2nd term.

6 Using the relevant row of Pascal's triangle, determine the coefficient of each term in the following expansion:

$$(x + m)^4 = \square x^4 m^0 + \square x^3 m^1 + \square x^2 m^2 + \square x^1 m^3 + \square x^0 m^4$$

Binomial theorem

7 Evaluate the following:

a $\binom{6}{5}$ **b** $\binom{14}{1}$ **c** $\binom{10}{9}$ **d** $\binom{8}{0}$

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8 Find the specified term for the following expansions:

a Fifth term of $(b+2)^7$.

b Fifth term of $(c+d)^9$.

c Fifth term of $(3u-v)^6$.

d Tenth term of $\left(u+\frac{1}{2}\right)^{12}$.

e Seventh term of $\left(\frac{3y}{2}-\frac{2}{3y}\right)^{10}$.

f Eleventh term of $\left(2x+y^2\right)^{13}$.

g Sixteenth term of $\left(x-y^5\right)^{19}$.

h Middle term of $\left(4x^2+3y^3\right)^8$.

i Middle term of $\left(-3x^{-3}+2y^{-2}\right)^4$.

j Constant term of $\left(2y-\frac{1}{y^3}\right)^8$.

9 The first three terms of $(x+y)^{10}$ are x^{10}, $10x^9y$ and $45x^8y^2$. Using the symmetry of the coefficients, write the last three terms of the expansion.

10 In the expansion of $(m+k)^n$, the coefficient of the 9th term is 45.

a Find the value of n.

b Find the 9th term.

11 A particular term in the expansion of $\left(3a^2+\frac{p}{b}\right)^4$ is $\frac{96a^2}{b^3}$, for some constant p. Find the value of p.

12 Consider the expansion of $(a-4)^5$. Will the 2nd term be positive or negative?

13 By considering $(x+y)^4=(x+y)(x+y)(x+y)(x+y)$, in how many ways can three factors of x and one factor of y be chosen from the expansion to form the term which contains x^3y^1? Write your answer in the form ${}^nC_r$.

14 Complete the following expansion:

$$(4p+3q)^3={}^3C_0\,(4p)^{\square}\,(3q)^0+{}^3C_1\,(4p)^{\square}\,(3q)^1+{}^3C_2\,(4p)^{\square}\,(3q)^2+{}^3C_3\,(4p)^{\square}\,(3q)^3$$

15 Consider the binomial $(4x+3y)^4$.

a State the first term in the expansion.

b State the last term in the expansion.

16 Consider the expansion of $(a+b)^5$.

a State the coefficient of the 3rd term in the form $\binom{n}{r}$.

b Evaluate the coefficient of the 3rd term.

c By considering the symmetry of $\binom{n}{r}$, which other term has the same coefficient as the 3rd term?

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17 Which two terms in the expansion of $(u+v)^{11}$ have a coefficient of $\binom{11}{9}$?

18 Find the coefficient of x^{17} in the expansion of $\left(3x^2+2x\right)^{11}$.

19 Use the binomial theorem to expand the following expressions:

a $(1+x)^4$ **b** $(2+b)^5$ **c** $(p+q)^5$ **d** $(c-d)^8$

e $(u+4v)^6$ **f** $(y+3)^4$ **g** $(ka-b)^6$ **h** $(y-5)^3$

i $\left(y-\frac{1}{3}\right)^3$ **j** $\left(2+\frac{y}{2}\right)^3$ **k** $(3y+2r)^3$ **l** $\left(a-\frac{1}{a}\right)^3$

m $\left(u^2+3v^2\right)^3$ **n** $\left(1+\frac{4}{y}\right)^3$

20 Find:

a The term in $(3+\sqrt{x})^{10}$ that contains x^4. **b** The term in $(c+d)^{10}$ that contains c^3.

c The term in $(u+v)^{11}$ that contains v^8. **d** The term in $(u+v)^{15}$ that contains u^5.

21 If the eighth and tenth terms in the expansion of $(x+y)^n$ have the same coefficients, find the value of n.

22 Find the coefficient of a^3b^6 in the expansion of $\left(2a-\frac{b}{2}\right)^9$.

23 Consider the expansion of $(1+x)^9$. State the coefficient of the 4th term in the form $\binom{n}{r}$.

24 Find the value of n such that the expansion of $(a+b)^n$ contains the term $84a^6b^3$.

25 Complete the following expansion by determining the missing power and binomial coefficient of the $(k+1)$th term:

$$(3-x)^8 = {}^8C_0 \times 3^8\,(-x)^0 + {}^8C_1 \times 3^7\,(-x)^1 + \ldots + \blacksquare \times 3^{8-k}\,(-x)^{\blacksquare} + \ldots + {}^8C_8 \times 3^0\,(-x)^8$$

26 Consider the binomial series:

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \ldots$$

a Use the first four terms to approximate $(1.03)^{-2}$ to the nearest thousandth.

b Use the first four terms to approximate $(1.05)^{\frac{3}{4}}$ to the nearest thousandth.

c Use the first five terms to approximate $\frac{1}{(1.06)^2}$ to the nearest thousandth.

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